

EXPERIENCE

Stripe

Principal Engineer, Infrastructure & Performance · 2019–Present

- Architected and led 12-person team delivering next-generation payment processing pipeline; reduced P99 latency by 340ms (from 450ms → 110ms) and increased throughput 2.8x (180K → 504K req/sec) serving \$2.2T+ annual payment volume
- Designed distributed consensus protocol for real-time settlement reconciliation across 18 global regions; achieved 99.9997% consistency (3 nines improvement) while cutting operational overhead by \$4.2M annually
- Mentored 5 senior engineers to principal-track roles; established incident command training program reducing MTTR by 62% and incident severity by 58% across 200+ microservices

Cloudflare

Senior Staff Engineer, Network Systems · 2016–2019

- Built HTTP/3 and QUIC optimization layer deployed across 280+ edge locations processing 40M req/sec; improved median latency 230ms → 85ms (63% reduction) and reduced packet loss recovery overhead by 45%
- Led platform reliability initiative improving global DNS resolution SLA from 99.95% → 99.999% (4-nines improvement); saved customers ~\$18M in potential downtime costs annually
- Championed adoption of eBPF-based network telemetry across infrastructure; cut observability costs by 67% (\$2.8M savings) while increasing metric resolution from 60s to 1s intervals

Shopify

Staff Engineer, Platform & Databases · 2013–2016

- Redesigned distributed sharding architecture for primary transactional database (900GB+, 250K queries/sec); achieved 4x improvement in query latency (from 280ms → 70ms) and enabled 5.2x capacity scaling without hardware upgrades
- Implemented real-time replication system for cross-datacenter failover; reduced RTO from 45 minutes → 4 minutes and RPO from 30s → <100ms, earning \$6.8M in SLA credit avoidance
- Managed platform infrastructure for 850K+ merchant stores; optimized resource allocation reducing cloud spend by \$3.1M YoY while maintaining 99.98% uptime SLA across containerized workloads

Google

PROJECTS

CacheMeld

Rust, C++, Redis, Memcached · 2020–Present

- Open-source distributed cache coherence framework with intelligent multi-tier invalidation; 4,200+ GitHub stars, 187K monthly npm downloads, and adopted by 40+ companies including 2 Fortune 500s
- Achieved 95th percentile cache hit rate 87.4% on real-world workloads vs. 71% for competitors; provided 4.2x memory efficiency through adaptive compression, reducing production cache footprint by \$800K annually for users

VectorFlow

Go, Python, TensorFlow, Protocol Buffers · 2018–2022

- Machine learning inference serving platform with dynamic batching; achieved 8,600+ GitHub stars and integrated into 12 production systems at Stripe, Shopify, and Figma
- Delivered 6.8x throughput improvement (from 420 → 2,856 inferences/sec per GPU) through model quantization and batch optimization; reduced inference latency from 142ms → 38ms while cutting GPU costs by 58%

RaftControl

Go, Protobuf, etcd, Kubernetes · 2019–2021

- Consensus algorithm testing and verification framework; 2,100+ GitHub stars with adoption in 18 production systems; featured in 3 academic papers on consensus correctness
- Reduced consensus failure detection time from 14s → 1.2s through improved leader election logic; generated 500K+ test scenarios covering edge cases, catching 47 previously unknown bugs in popular implementations

EXTRACURRICULAR

Strange Loop Conference & QCon

Technical Speaker & Program Committee Member · 2017–Present

- Delivered 6 keynotes and technical talks on distributed systems, concurrency, and performance optimization with combined audience reach of 8,400+ engineers; talks achieved 4.7/5.0 average speaker rating
- Served on program committee for 4 conferences selecting 180+ talks from 2,100+ submissions; mentored 12 first-time speakers resulting in 9 accepted presentations and fostering emerging voices in systems community

Open Source Systems Mentorship Initiative

Principal Mentor & Advisor · 2016–Present

- Actively mentor 7 open-source projects across Rust ecosystem; 4 mentees

EDUCATION

ETH Zurich (Swiss Federal Institute of Technology)

MSc Computer Science · 2010–2012

- Thesis on distributed systems optimization achieved 9.2/10 (top 3% cohort); published 2 peer-reviewed papers on lock-free concurrent data structures in IEEE Transactions on Parallel and Distributed Systems
- Led Systems Engineering Club with 150+ members; organized 8 technical workshops on concurrency patterns, attracting speakers from Google and Microsoft

University of Warsaw

BSc Mathematics & Computer Science · 2006–2010

- Graduated summa cum laude with 3.95/4.0 GPA; awarded Faculty Excellence Award for contributions to algorithmic research
- Conducted undergraduate research on randomized algorithms under Prof. Marek Cygan; co-authored 1 publication in competitive programming domain

SKILLS

Languages

Go Rust C++ Python Java
SQL Bash JavaScript

Technologies

Kubernetes Distributed Systems
Consensus Algorithms (Raft Paxos)
Protocol Buffers gRPC REST APIs
Event Streaming (Kafka) PostgreSQL
DynamoDB Redis
Message Queues (RabbitMQ)
TensorFlow Docker Terraform
etcd Datadog Prometheus

Tools

Git Linux Docker Kubernetes
Terraform CloudFormation
GitHub Actions Jenkins Prometheus
Grafana Datadog PagerDuty
VirtualBox GCP AWS Azure
Bazel

- Developed high-performance RPC framework for MapReduce clusters processing 8 exabytes/day; cut request serialization overhead by 73% through custom binary protocol and reduced tail latency (p99) from 320ms → 80ms
- Owned cluster resource scheduler serving 14K compute nodes across 6 datacenters; improved resource utilization by 31% (\$12M annual savings) and reduced job startup time by 52%
- Co-authored 2 internal papers on distributed tracing and adaptive batching; knowledge transferred to 28 engineers across 4 teams reducing incident resolution time by 44%

promoted to core maintainer roles within 18 months; mentored codebase now maintains 99.2% test coverage with zero critical CVEs in 3-year period

- Authored comprehensive distributed systems and performance optimization guide (100K+ GitHub clones, 340K readers); established weekly office hours attended by 45+ engineers, providing technical guidance on architecture decisions and code reviews